

Mid-Project Report

Design and Development of an AI-Powered Document Assistant Based on Retrieval-Augmented Generation (RAG)

Student: Rim Bousta

Supervisor: Othmane Goulida and Mahmoud Attiff

Company: *Omnishore, Groupe Medtech.*

Internship Period: *2 months (1 month through)*

1. Introduction

As part of this internship, the objective is to design and develop an intelligent document assistant capable of answering users' questions by retrieving information exclusively from a collection of documents using Artificial Intelligence. (for testing so far, I used 11 homogeneous documents)

The application is based on the **Retrieval-Augmented Generation (RAG)** architecture, which combines semantic search with a Large Language Model (LLM) to generate reliable answers supported by references extracted just from the source documents.

This report presents the progress achieved so far, the current implementation, the experiments conducted, and the next development steps (les axes d'amélioration).

2. Project Objectives

The main objectives of the project are:

- Automatically extract information from PDF documents.
- Convert documents into semantic vector representations.
- Store document embeddings in a vector database.
- Retrieve the most relevant document passages according to a user's question.
- Generate natural language answers using a Large Language Model.
- Provide citations and sources to ensure answer traceability and reduce hallucinations.

3. Current System Architecture

The implemented pipeline currently consists of the following stages:

1. PDF extraction using **Docling**
2. Semantic chunking of extracted text (At first, I started with static chunking but I moved after to semantic chunking)
3. Embedding generation using **BGE-M3**
4. Vector indexing with **Qdrant**
5. Semantic retrieval of relevant chunks
6. Reranking using **BGE-Reranker** (I started first with the larger version bge-reranker-m3-v2 but then moved to a lighter version bge-reranker-base)
7. Answer generation with **Mistral 7B (Ollama)**

The system returns both the generated answer and the document excerpts used to produce it.

4. Work Completed

At this stage, the following functionalities have been successfully implemented:

- Complete RAG pipeline.
- Automatic PDF ingestion and processing.
- Semantic chunk generation.
- Vector indexing in Qdrant.
- Semantic retrieval.
- Reranking of retrieved passages.
- Local answer generation using Mistral 7B.
- Streamlit-based graphical user interface.
- Source citation display.
- Execution time monitoring for each pipeline stage.
- Interactive document upload and question-answer interface.

The application is now fully operational from document upload to answer generation, but some concerns around the quality of answers were raised.

5. Performance Optimization

Several experiments were conducted to improve the overall performance of the system. (taking into account the hardware limitations. I conducted these experiments in my own personal computer that has the following configurations and no GPU.)

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\l>wmic computersystem get TotalPhysicalMemory
TotalPhysicalMemory
16985653248

C:\Users\l>wmic cpu get name, numberofcores
Name                                NumberOfCores
Intel(R) Core(TM) i7-8665U CPU @ 1.90GHz 4

C:\Users\l>
```

The evaluated parameters include:

- Reranker model selection.
- Number of retrieved chunks.
- Chunk size.
- Number of chunks provided to the language model.

A benchmarking process was performed to compare different configurations.

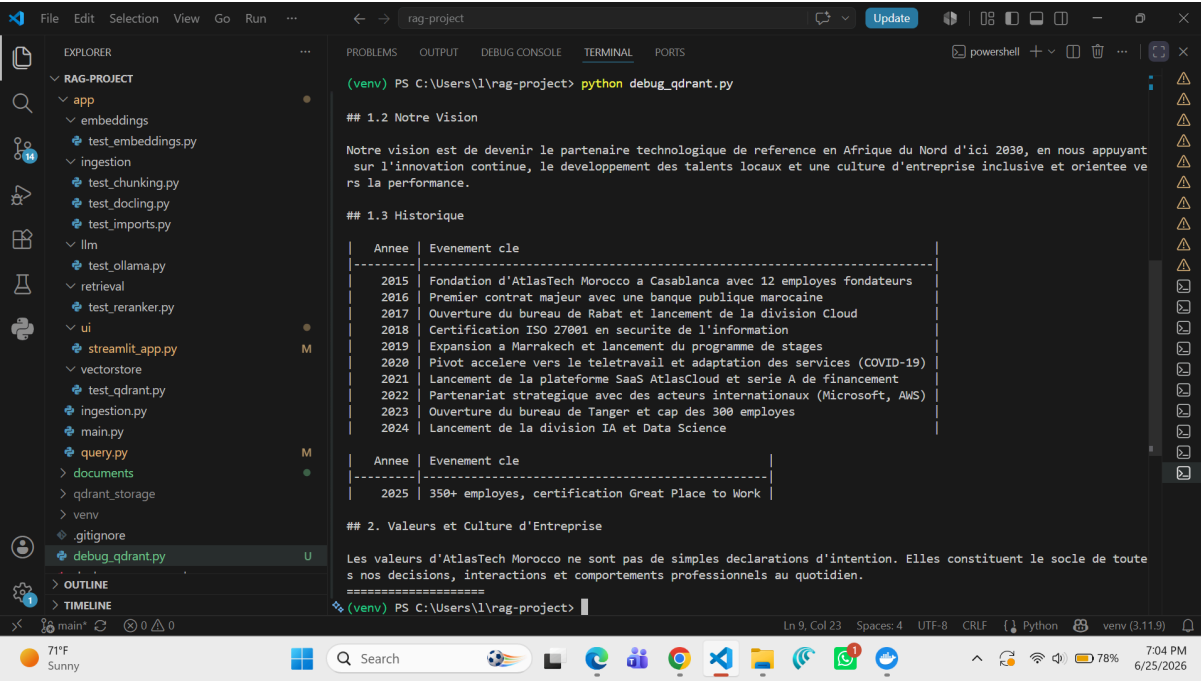
Configuration	Average Response Time	Quality
Initial configuration	around 282 seconds	Very Good
Optimized configuration	around 78 seconds	Good and factually reliable

The generated answers were manually compared with the original documents. No hallucinations were observed during the conducted tests, while the execution time was reduced by approximately **3.6×**.

6. Current Challenges

Although the system performs well on most document-related questions, a limitation has been identified when retrieving information contained in structured tables.

For example, the **"History"** section of one document is correctly extracted and stored in the vector database but is not consistently retrieved during semantic search, even when relevant queries are asked. (such as “Quel est l’historique de la société ?”). As shown below, it is extracted and appears in both the output terminal during debugging and the UI in the sources yet it is not visible to Mistral:



1.3 Historique

Annee	Evenement cle
2015	Fondation d'AtlasTech Morocco a Casablanca avec 12 employes fondateurs
2016	Premier contrat majeur avec une banque publique marocaine
2017	Ouverture du bureau de Rabat et lancement de la division Cloud
2018	Certification ISO 27001 en securite de l'information
2019	Expansion a Marrakech et lancement du programme de stages
2020	Pivote accelere vers le teletravail et adaptation des services (COVID-19)
2021	Lancement de la plateforme SaaS AtlasCloud et serie A de financement
2022	Partenariat strategique avec des acteurs internationaux (Microsoft, AWS)
2023	Ouverture du bureau de Tanger et cap des 300 employes
2024	Lancement de la division IA et Data Science

Annee	Evenement cle
2025	350+ employes, certification Great Place to Work

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SOURCE 1

score 0.003

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AtlasTech Morocco

Manuel de l'Employe

Version 2.0 - Departement des Ressources Humaines Janvier 2025

1. Presentation de l'Entreprise

AtlasTech Morocco est une societe technologique de premier plan implantee a Casablanca, avec des bureaux regionaux a Rabat, Marrakech et Tanger. Fondée en 2015, l'entreprise s'est rapidement imposee comme un acteur cle du secteur numerique marocain, proposant des solutions innovantes en matiere de transformation digitale, de cybersécurité, de cloud computing et de developpement logiciel sur mesure.

Avec une equipe de plus de 350 professionnels qualifiés et une clientele diversifiée couvrant les secteurs bancaire, industriel, sante, education et services publics, AtlasTech Morocco incarne l'excellence technologique au service du developpement economique du Maroc et de la region MENA.

Notre siege social est situe au Technopark de Casablanca, un ecosysteme d'innovation qui nous permet de rester a la pointe des evolutions technologiques mondiales tout en ancrant notre identite dans le tissu economique local.

1.1 Notre Mission

La mission d'AtlasTech Morocco est d'accelerier la transformation numerique des organisations en leur fournissant des solutions technologiques fiables, securisees et adaptees a leurs besoins specifiques. Nous croyons que la technologie, lorsqu'elle est deployee de maniere strategique, peut creer une valeur durable pour nos clients, nos employes et la societe dans son ensemble.

1.2 Notre Vision

Notre vision est de devenir le partenaire technologique de reference en Afrique du Nord d'ici 2030, en nous appuyant sur l'innovation continue, le developpement des talents locaux et une culture d'entreprise inclusive et orientee vers la performance.

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2. Valeurs et Culture d'Entreprise

Les valeurs d'AtlasTech Morocco ne sont pas de simples declarations d'intention. Elles constituent le socle de toutes nos decisions, interactions et comportements professionnels au quotidien. ...

2

SOURCE 2

Doc2_AtlasTech_Manuel_Employe.pdf

score 0.000

), des plans d'evacuation sont affiches dans chaque zone et des exercices reguliers sont organises.

Several approaches have already been investigated, including:

- Semantic chunking.
- Larger chunk sizes.
- Metadata enrichment.
- Keyword-based fallback retrieval.

The issue remains under investigation and appears to be related to the retrieval/reranking stage rather than the extraction process.

Additionally, response generation remains relatively slow due to the local execution of the Mistral 7B model on CPU without GPU acceleration.

7. Next Steps

The next phase of the project will focus on extending both the functionality and the robustness of the assistant.

The planned developments include:

- Determining the minimum hardware configuration required to achieve acceptable response times.
- Document version management. (the versionism problem mentionned in early deliverables state_of_the_art)
- Conversation history storage.

- User feedback collection (usefulness rating).
- Administrator notifications for unanswered questions.
- Integration with a Document Management System (DMS), such as Alfresco.
- OCR support for scanned documents.
- Usage statistics and monitoring dashboard.
- Exploration of middleware integration with existing enterprise systems.

8. Conclusion

At this stage of the internship, a fully functional AI-powered document assistant has been successfully developed.

The complete RAG pipeline has been implemented and validated through several experiments and performance benchmarks. The system is already capable of answering questions based on uploaded documents while providing supporting citations.

The remaining work will focus on improving retrieval quality for specific document structures, extending the supported document corpus, and integrating additional enterprise-oriented features to make the solution more robust and scalable.